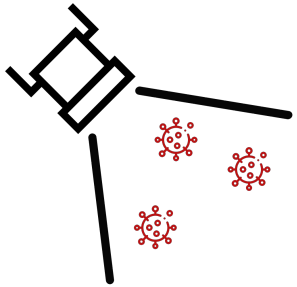


FloodLAMP
Biotechnologies, PBC

Delivering the testing our world needs
For COVID-19 and beyond

Randy True, Founder and CEO | randy@floodlamp.bio

Addressing Current Vulnerabilities

Antigen Tests are at Risk of Failure with New Variants

- Preliminary observational reports of antigen test failures in multiple outbreaks (BinaxNOW)
- Even if current antigen tests prove to detect Omicron, we're at risk of that not being the case with the next VOC

Multi-Gene Molecular Tests are Resilient to New Variants

- Our test combines primer sets for 3 genes: ORF1ab, N2 and E1
- LAMP uses 6 primers and is highly tolerant to mismatches (see recent NEB preprint analyzing our primers)

Reagent Based Molecular Tests are Adaptable

- Primers can be swapped out quickly and cheaply, with all other test components remaining the same
- Primers can be produced in large volumes and distributed quickly (10's of M of reactions per production run)

FloodLAMP can deliver Prepositioned Assets

- For distributed, rapid response surge capability
- Our systems comprise ordinary low cost lab equipment and offer best-in-class capacity per cost
- We deliver custom-designed wraparound components (digital tools, QA, training) that enable rapid scaling

Bypassing Typical Limitations

Not limited by CLIA labs

- Non-Dx "surveillance" screening permitted by FDA and CMS but potential has not yet been realized at scale.
- Huge advantage in flexibility for deployment to widely distributed bare-bones test processing sites, which minimizes turnaround time and give highly valued local control and resiliency.

Not limited by need for certified clinical lab techs

- Staffing shortages in the testing field are stifling. We can bring on and train new test operators very quickly while simultaneously ensuring test safety, reliably, and traceability.

Not limited by upfront capital equipment

- $2K of standard, non-proprietary equipment gives 20K/day screening capacity.
- Sites can typically operate far under max capacity, and then quickly surge with addition of extra labor.

Not limited by centralized manufacturing and sole sources

- Our test use standard consumables and has reagent supplier redundancy.

Filling a Gap in the Testing Landscape

Current testing paradigms cannot stem a pandemic caused by an asymptomatically transmitted, aerosolized pathogen.

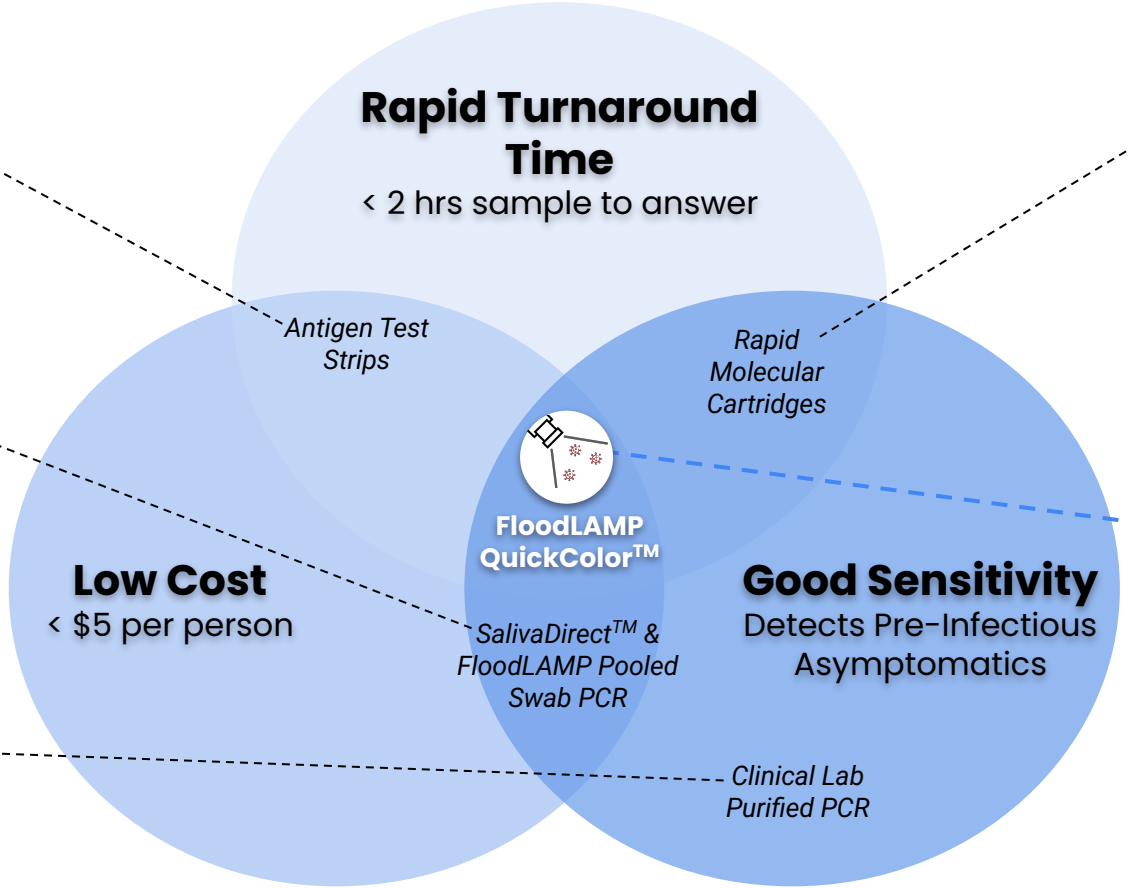
- Scaling a single test up to very high levels offers **opportunities for innovation** in sampling, test chemistry, and program configuration.
- A "**Point of Need**" modality of testing is needed that combines the scalability and low cost of liquid reagent processing with the ease, flexibility and decentralized nature of POC/At-Home.
- **Public Health processing sites** can fill the gap between CLIA labs and DIY at home tests.
- Want this new capability to be **additive** and not cannibalize the clinical diagnostic infrastructure.

Performance in All Dimensions

People want higher confidence for both positive and negative results.

High upfront capital cost for PCR instruments limit replicability. These 2 tests have potential to drive down PCR costs.

Long turnaround renders it unsuitable for large scale screening - schools and workplaces need to know quickly if someone is infected.



Will help usher in era of at-home concierge diagnostics but simply too constrained by manufacturing and too expensive for public health screening.

- **Local**
- **Fast**
- **Cheap**
- **Molecular**
- **Flexible**
- **Resilient**

FLOODLAMP PLATFORM

Applications

- COVID Response
- Emerging Threats (Pandemic Prep)
- TB Zikka
- Influenza STD's Cancer

Core Modules

PHYSICAL

- Reagent Test Kits
- Collection Kits
- Standard Equipment
- Standard Consumables

DIGITAL

- App
- Admin Portal
- Training Program
- Quality Management System

Network

- EMS & Municipalities
- Schools and Businesses
- U.S. Federal Agencies (BARDA, ASPR, CDC, DoD, FEMA)
- International Governments

Products

- All-in Test Kits
- Sample Processing Sites (Labs)
- Service, Support, Consulting
- Testing Program Management

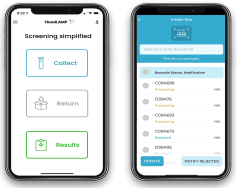
Our Unique Solution

Standalone Components and Fully Integrated Program

Pooled Swab Collection

At-home Family Pooling

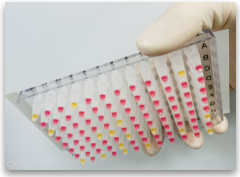
- Up to 4 nasal swabs per sample tube
- Upfront at-home and on-site pooling makes lab processing far more efficient



FloodLAMP Mobile App

Decentralized Accessioning

- Supports both self and sponsored collection
- Results can be reported directly to participants and administrators
- Lab staff and program admin interfaces



FloodLAMP QuickColor™ Test

Molecular Performance

- RT-LAMP "PCR-like" molecular amplification
- No instruments to run test or analyze results
- Low cost and easy to run
- Results in 45 minutes with only 15 min hands-on

Turnkey Complete Packages

1000's Samples/Day Anywhere

- Everything needed to bringup program from empty room with 3 tables
- Comprises calibrated standard equipment, supplies, consumables, and test reagent kits (50+ SKUs, many pre-prepped for ease of use)

Quality System

Streamlined traceability

- In development with top QMS consulting group
- Covers suppliers, manufacturing, and test processing

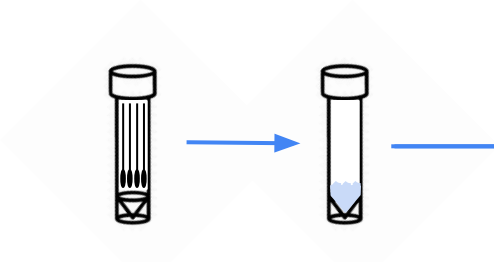
Training Program

Remote certification

- In development with top learning experience consulting group
- Assessment for understanding and compliance
- Enables rapid growth while maintaining quality

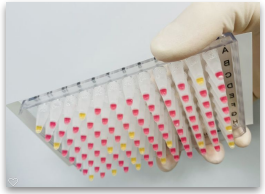
Core Assay Technology

FloodLAMP has **fully validated** 2 complementary tests that are best-of-breed with EUAs submitted to the FDA. Pooled home collection kit was also **submitted for interactive review**. The test workflow and collection kit has been designed to **expand to other disease targets** in the future.



Streamlined Sample Prep

- Upfront swab pooling
- Highly scalable, integrated processing
- Same sample for both tests



Visual Readout

QuickColor™ LAMP Test*

- High sensitivity (90%)
- Ultra-high throughput
- Ideal for serial screening
- Uniquely scales without capital intensive instruments

**Licensed from Harvard Medical School (Rabe Cepko - "SARS-CoV-2 detection using isothermal amplification and a rapid, inexpensive protocol for sample inactivation and purification" PNAS 09-29-2020, MGH Clinical Validation Preprint)*

Standard RT-qPCR instrument

EasyPCR™ Test

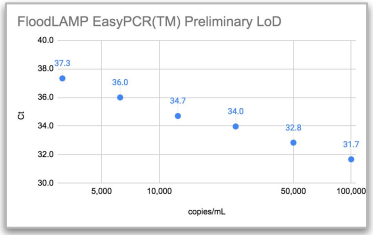
- Very high sensitivity (98%)
- Medium throughput (1.5 hrs/94)
- Ideal for diagnostics and confirmatory reflex

Clinical Evaluation Data

Clinical evaluation performed by the Stanford CLIA Lab, with excellent results and praise on the "really straightforward" protocol.

EasyPCR™ Test

- 3 copies/µl LoD
- 98% sensitivity (PPA 39/40)
- 100% Specificity (40/40)
- No false positives



Gamma inactivated cell lysate from BEI spiked into raw clinical negative sample

FloodLAMP SwabDirect PCR Result	Comparator Positive	Comparator Negative	Total
Positive	39	0	39
Negative	1	40	41
Invalid	0	0	0
Total	40	40	80
Positive Agreement	97.5% (39/40) 95% CI: 86.8% to 99.9%		
Negative Agreement	100% (40/40) 95% CI: 91.2% to 100%		

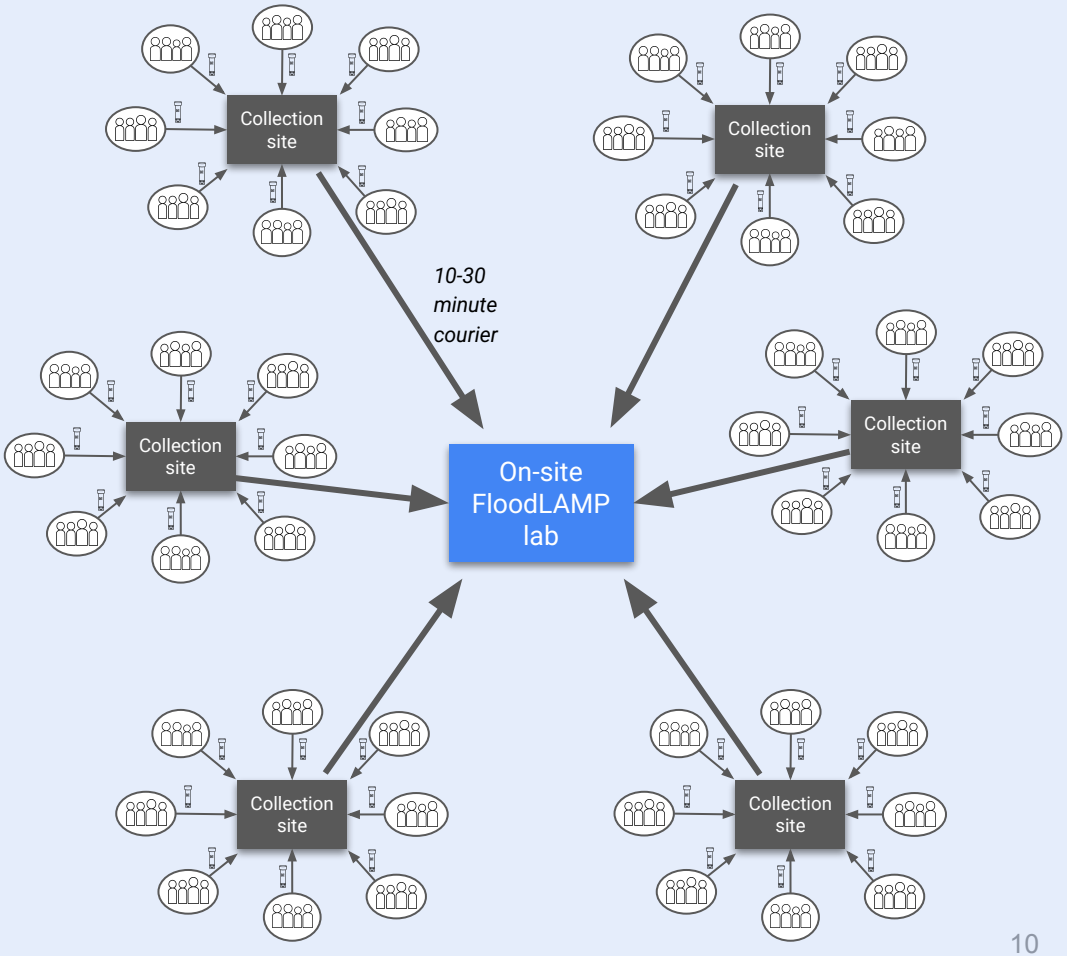
QuickColor™ LAMP Test

- 12 copies/µl LoD
- 90% Sensitivity (PPA 36/40)
- Missed positives only high Ct (>36 with direct PCR)
- 100% Specificity (40/40)
- No false positives

FloodLAMP QuickColor Test Result	Comparator Positive	Comparator Negative	Total
Positive	36	0	36
Negative	4	40	44
Total	40	40	80
Positive Agreement	90.0% (36/40) 95% CI: 76.3% to 97.2%		
Negative Agreement	100% (40/40) 95% CI: 91.2% to 100%		

Source of Specimens: Stanford COVID-19 Clinical Testing Program
Specimen Type: Anterior Nares Swab in PBS, previously tested and frozen
Comparator Test: Hologic Panther Fusion SARS-CoV-2 Assay and Hologic Panther Aptima SARS-CoV-2 Assay

Program mechanics

EMS/Employer program

- On-site employee pooled collection
- Multiple collection points
- Courier service to lab (10-30 minutes)
- On-site lab
- Case study of hundreds of employees screened daily by one tester in 2 hrs
- Cost $2 per person
- Flexibility for follow up testing of families and contacts
- Case study of asymptomatic town manager testing positive, and using FloodLAMP to immediately test his family, picking up 2 asymptomatic positive kids and stopping exposure in their schools